ECERC Limitation 2: Class Imbalance on MELD

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Summary

None of our four attempted fixes reached the pre-registered goal of +1.0 macro-F1 for ECERC on MELD. The closest was fusing ECERC with a fine-tuned RoBERTa-large text model. It lifted Fear F1 from 26.1 to 29.5, but macro-F1 rose only 0.27 points (95% CI −1.92 to +2.29).

Limitation 2 of ECERC (Zhang & Tan, ACL 2025) is that rare emotions, chiefly Fear and Disgust, are lost to Neutral and to related emotions. We ran one diagnostic and four attempted fixes: at the decision rule, inside ECERC's training, and by fusing ECERC with two stronger text models. Every choice was made on the dev set, and a dev gate stopped two experiments, so the test set was opened only twice.

The evidence points to three causes. Rare emotions overlap with related ones in MELD's labels (annotator agreement κ = 0.43), the frozen input features carry little rare-class signal, and any model trained on MELD's skewed labels inherits the skew. A real fix likely needs richer audio-visual features, more or relabeled rare-class data, or end-to-end encoder training, which were beyond our data and free-GPU budget.

1. Background: ECERC, MELD and Limitation 2

ECERC recognizes MELD's rare emotions far worse than its common ones: 26.12 test F1 for Fear and 31.43 for Disgust, against 79.80 for Neutral. Its paper reports 67.32% accuracy and 66.46 weighted F1, averaged over three runs.

The model. ECERC is an emotion-recognition-in-conversation (ERC) model: it labels every utterance in a dialogue with one of seven emotions. It runs on frozen, precomputed features: RoBERTa-large text features fine-tuned on MELD's training labels (the COSMIC recipe), 300-d openSMILE audio and 342-d DenseFace visual features. Four gated attention branches model emotion and event causes from the same and other speakers, and one linear layer classifies.

The data. MELD has 13,708 utterances from 1,432 dialogues of the TV series *Friends*, each labeled by three annotators (Fleiss' κ = 0.43). Neutral is 47% of the training set, while Fear and Disgust are 2.7% each, a 17:1 ratio. The dev set has only 40 Fear and 22 Disgust utterances.

Emotion	Train	Dev	Test
Neutral	4,710	470	1,256
Joy	1,743	163	402
Surprise	1,205	150	281
Anger	1,109	153	345
Sadness	683	111	208
Disgust	271	22	68
Fear	268	40	50
Total	9,989	1,109	2,610

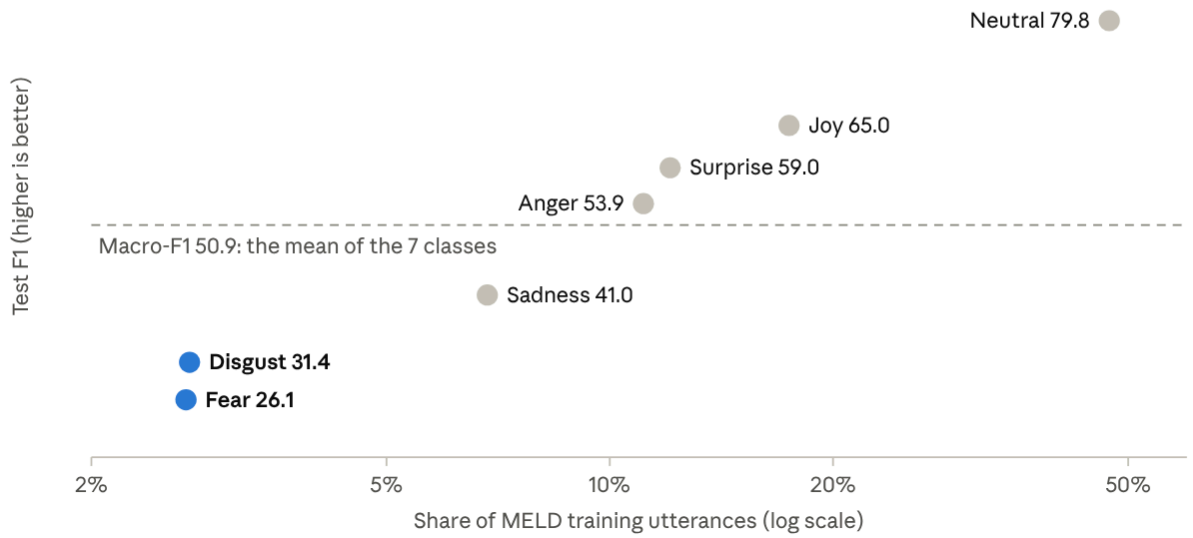
The limitation. The paper names "misclassification of closely related categories, as well as the imbalance in sample sizes" as a limitation. It blames "excessive Neutral training samples" but does not test that cause. In its own confusion matrix, 32% of Fear goes to Neutral and 46% to other emotions; Disgust loses 35% to Neutral and 26% to Anger.

The metric. The paper reports only accuracy and weighted F1, which Neutral dominates. We use macro-F1, the plain mean of the seven class F1 scores, so each emotion counts equally. From the paper's Table 3, ECERC's macro-F1 is 50.87.

An earlier attempt. Before this study, class-weighted loss and oversampling were tried with 3 seeds, selected on dev weighted F1. They raised Fear recall but cut its precision, with no F1 gain. That study was too small to be conclusive and is reported only as context.

Test F1 falls in rank order of class frequency, down to 26.1 for Fear

Each dot is one emotion (ECERC on MELD). Blue: Fear and Disgust, this study's targets.



ECERC test F1 per class: mean of the three released checkpoints, reproduced in experiment 1 and equal to the paper's Table 3. Class shares: MELD training split.

The problem is not specific to ECERC: the seven baselines in the ECERC paper score 0 to 22.2 F1 on Fear and 0 to 29.8 on Disgust.

2. Approach: pre-registered criteria and a sealed test set

Every experiment followed one rule: no decision could depend on test labels. Methods, thresholds, weights and checkpoints were chosen on the dev set, and each experiment's rules were written down before it ran.

Research question. Can ECERC's macro-F1 on MELD be raised without hurting its overall accuracy? An improvement counted only if it met all of these pre-registered criteria on the test set:

1. A macro-F1 gain of at least +1.0 point, with a 95% confidence interval (CI) above zero.
2. Both Fear F1 and Disgust F1 improve.
3. Accuracy drops by no more than 1.0 point, and weighted F1 by no more than 0.5 point.
4. No class's F1 drops by more than 2.0 points.
5. The gain is consistent: positive for every checkpoint (experiment 1), or for at least 70% of seeds or model pairs (experiments 3 to 5).

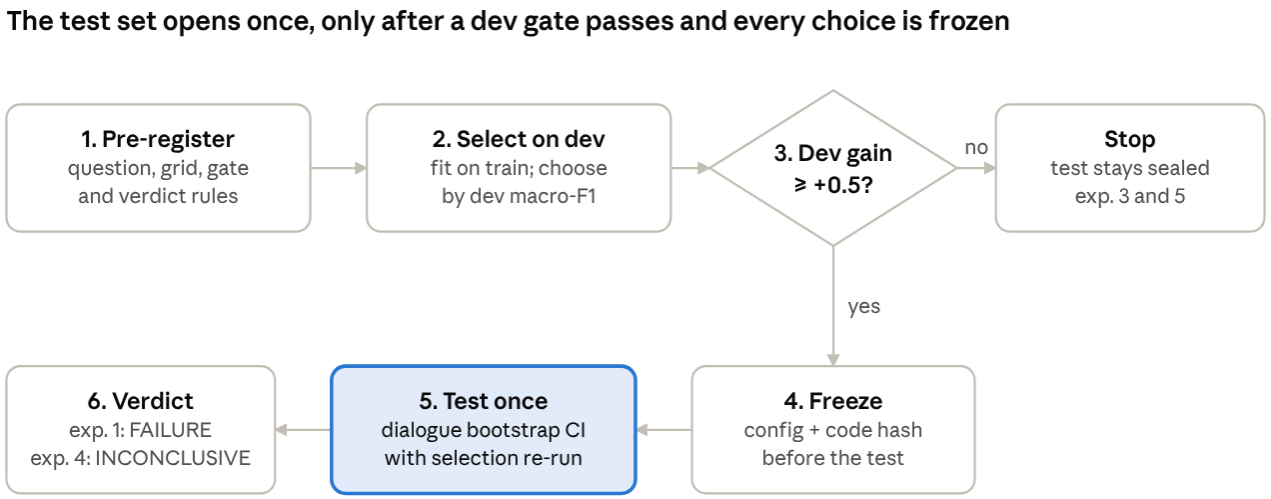
Verdicts were applied mechanically; the first matching row wins.

Verdict	Rule
SUCCESS	All five criteria met
TRADE-OFF	Criteria 1, 2 and 5 met, but a guardrail (3 or 4) fails
PARTIAL	Criterion 1 met, but Fear, Disgust or consistency fails
FAILURE	The CI's upper bound is below +1.0, so the target is excluded
INCONCLUSIVE	Anything else

Safeguards.

- **Priors from training labels only.** The repository's hardcoded class weights include test labels, so they were never used.
- **Dev-only selection.** Every threshold, fusion weight, candidate and epoch was chosen by mean dev macro-F1.
- **A dev gate.** In experiments 3 to 5, the test ran only if the dev gain reached +0.5. In experiments 4 and 5 that gain was cross-fitted: settings chosen on four folds of dev dialogues, scored on the fifth.
- **A sealed, single test pass.** All decisions were written to a frozen config together with the code's SHA-256 hash. The test stage refused to run if either had changed, and it ran once.
- **Honest intervals.** CIs come from a dialogue-level bootstrap (10,000 replicates) that repeats the dev selection inside every replicate, so they include selection noise.
- **Fidelity.** Our pipeline reproduced the original evaluation code exactly on dev, and the baseline's test scores matched the paper's Table 3.
- **Leak tests.** Each notebook was first run on synthetic data, with a guard that fails if any test label is read before the test stage.

Baseline and compute. The baseline is the authors' three released MELD checkpoints (C1 to C3), whose mean reproduces the paper's Table 3; experiment 3, which retrains ECERC, uses our own retraining instead. Training ran on free Tesla T4 GPUs on Colab and Kaggle. Code and tests were developed on an 8 GB laptop with no GPU.



Decision flow of experiments 1, 3, 4 and 5 · 6 steps, 1 gate

Experiments 3 and 5 stopped at the gate, so only experiments 1 and 4 opened the test set. Experiment 1 had no gate; a pre-test power check showed it could detect the target. Experiment 2 never needed the test set.

3. Results at a glance

No experiment met the success criteria. Only experiment 4 improved a rare class on test (Fear), and its overall gain was too small to confirm.

#	Experiment	Question	Key result	Verdict	Test set used
1	Post-hoc logit adjustment	Is it a decision-threshold problem?	Δ macro-F1 −0.24 [−1.06, +0.68]	FAILURE: target excluded	Yes, first look
2	Feature-leak check	Do the text features memorize the training labels?	Held-out gap +7.6 points; the rule needed 15	NOT CONFIRMED	No
3	Training-time changes	Can changes to ECERC's own training fix it?	Best dev gain +0.19; the gate needed +0.5	Stopped at the gate	No
4	ECERC + RoBERTa-large text expert	Does an end-to-end text model add rare-class evidence?	Δ macro-F1 +0.27 [−1.92, +2.29]; Fear F1 +3.40	INCONCLUSIVE	Yes, second look
5	ECERC + Qwen2.5-7B LLM expert	Does a much larger language model add more?	Cross-fitted dev gain −0.98; the gate needed +0.5	Stopped at the gate	No

Δ is the change against the released ECERC checkpoints, in macro-F1 points; brackets are 95% CIs.

4. Audit of the released code and checkpoints

The released code differs from the paper in ways that matter for class imbalance: a different loss, test labels inside the class constants, and checkpoint selection that favors Neutral. We read the code and checkpoints before any experiment.

Area	What we found	Why it matters
Loss	The paper describes cross-entropy (its Eq. 16). The MELD code uses a focal loss with $\gamma = 1$ whose $(1 - p)$ factor is detached from the gradient. IEMOCAP uses inverse-frequency-weighted cross-entropy, always on.	The MELD baseline already contains a mild rare-class correction, so a focal or weighted loss is not a new fix.
Class-frequency constants	The hardcoded MELD proportions equal the train + dev + test proportions to 9 decimals (Fear: 0.026116137 = 358 / 13,708). They are off by default for MELD. IEMOCAP's constants, on by default, also include its test session.	Any class weighting that uses them leaks test labels. We recomputed priors from training labels only.
Checkpoint selection	The best epoch is picked by dev weighted F1 (or dev loss). The training script also prints test scores every epoch and saves no model.	On dev, weighted F1 counts Neutral 12× more than Fear, so selection favors Neutral-leaning epochs. Printed test scores invite test-based choices.
Released checkpoints	Three MELD checkpoints on HuggingFace (zt-ai/ECERC): C1 (seed 0, chosen by dev F1), C2 (seed 2613, dev F1) and C3 (seed 659, dev loss). Their mean test scores equal Table 3 exactly.	The paper's numbers are the average of these three, so we used them as the baseline.

Area	What we found	Why it matters
Training regime	Adam at learning rate 1e-5 for about 1,320 steps lets each weight move at most about 0.013. LayerNorm gains moved at most 0.011, and the classifier biases stayed near their initial values (Neutral +0.003, Fear +0.017).	The Neutral preference is not stored in the classifier head, so bias or threshold corrections have little to undo.
Feature order	The loader returns visual features before audio, but inference names them the other way round. The 300-d "audio" input holds the first 300 visual dimensions; the "visual" input holds the last 42 visual dimensions plus all the audio.	We left this unchanged, because the released weights were trained this way.
Reproducibility	CPU runs crash, because attention masks are built only on CUDA. The text-feature extraction code is not released (GitHub issue #2 has no reply).	We added a small CPU shim for local tests. The features cannot be rebuilt.

5. Experiment 1: post-hoc logit adjustment

Shifting ECERC's decisions toward rare classes did not help. Macro-F1 changed by −0.24 points (95% CI −1.06 to +0.68), so the +1.0 target is excluded.

Idea. If Neutral wins because of its large prior, subtracting $\tau \cdot \log \pi$ from the log-probabilities should recover rare classes without retraining. Here π is the training class frequency, and $\tau = 1$ is the theoretical correction for balanced error (logit adjustment, Menon et al. 2021).

Setup. We used the three released checkpoints, unchanged. τ was chosen on dev from 0 to 2.0 in steps of 0.1; the best was $\tau^* = 0.1$. Two controls ran alongside: a penalty on Neutral only, and the full correction $\tau = 1$. Before the test, a power analysis showed that a 0.91-point effect would be detected 80% of the time, so a null result is informative.

Arm	Δ macro-F1 vs ECERC [95% CI]
Logit adjustment, $\tau^* = 0.1$ (primary)	−0.24 [−1.06, +0.68]

Arm	Δ macro-F1 vs ECERC [95% CI]
Neutral-only penalty (control)	−0.10 [−0.81, +0.58]
Full prior correction, τ = 1	−6.63 [−9.01, −4.16]

The primary arm changed Fear F1 by −1.09 and Disgust F1 by +0.06. Accuracy fell 0.59 points and weighted F1 0.30. Per checkpoint, the change was −0.60, +0.50 and −0.63.

What it showed.

- **The extra rare-class predictions are mostly wrong.** Nineteen extra Fear predictions gave one correct one: 5% precision at the margin, against 34% overall. Of 129 utterances moved away from Neutral, 70 really were Neutral.
- **Rare classes are confused with related emotions as much as with Neutral.** On dev, Fear goes to Neutral in 33% of cases and to Surprise, Sadness or Anger in 38%. Pairwise AUROC is 0.95 for Fear vs Neutral, but 0.84 to 0.87 for Fear vs those emotions, and 0.85 for Disgust vs Anger.
- **The errors are systematic.** Of 31 dev Fear items missed by any checkpoint, 30 are missed by all three; for Disgust, 14 of 15. More seeds or an ensemble of ECERC models would not recover them.
- **No context-copying failure.** Errors repeat the previous utterance's label in 25% of cases, below the 39% rate at which labels truly repeat.
- **ECERC underfits rare classes even on its training data.** Its Fear recall on the training set is 31%, and its training macro-F1 is 63.5 against 54 on dev.

Conclusion. Limitation 2 is not a misplaced decision threshold. The "Neutral attraction" the paper describes is not an output bias that a prior correction can remove.

6. Experiment 2: feature-leak check

The text features do not memorize the training labels enough to explain Limitation 2. The train-versus-dev gap was +7.6 macro-F1 points, below the pre-registered leak threshold of 15.

Idea. In experiment 1, a linear probe on ECERC's text features scored 87.7 macro-F1 on the training data it was fitted to, but 41.3 on dev. The text encoder was fine-tuned on those same training utterances, so their features might encode their labels. ECERC would then learn shortcuts that vanish on new data.

Setup. Logistic-regression probes were fitted on four fifths of the training dialogues and scored on the held-out fifth and on dev. The gap is held-out-train minus dev macro-F1. Plain RoBERTa features, never fine-tuned on MELD, serve as the control. A leak counted

only if the gap was at least 15 points and at least 10 above the control's, with a CI above zero. The test split was never read.

Features	Fitted train	Held-out train	Dev	Gap [95% CI]
Fine-tuned RoBERTa text (ECERC's input)	91.9	47.9	40.3	+7.6 [+4.4, +11.2]
Plain RoBERTa text (control)	75.9	36.8	36.8	+0.0 [−3.1, +3.3]
Audio (openSMILE)	36.4	29.6	22.4	+7.2 [+4.8, +10.0]
Visual (DenseFace)	26.4	14.0	14.0	−0.0 [−1.7, +1.7]

Values are probe macro-F1 on each set. A 1-nearest-neighbour check agreed: 46.3 on train against 42.1 on dev for the fine-tuned text features, and flat for the others.

What it showed.

- **The big drop in experiment 1 was ordinary probe overfitting.** On training dialogues it had not seen, the probe scored 47.9, close to dev.
- **A small train-side advantage is real.** It appears only in features built with label supervision: the fine-tuned text (+7.6) and the audio, whose dimensions were picked by supervised feature selection (+7.2). It is a secondary factor, not the main cause.
- **Rare-class signal is weak even within the training data.** On unseen training dialogues, the text features find only 24% of Fear utterances; on dev, 11%.

Conclusion. The features transfer as expected. The bottleneck is genuine overlap between emotions, plus label ambiguity, not a leak.

7. Experiment 3: training-time changes to ECERC

No change to how ECERC is trained moved dev macro-F1 beyond seed noise. The best variant gained +0.19 points against a +0.5 gate, so the test set was not used.

Idea. If the features hold the signal but ECERC fails to use it, changing the training should help. We tried richer text inputs, a loss that pushes confusable emotions apart, extra sentiment supervision and a higher learning rate.

Setup. ECERC's architecture, data loader and loss were unchanged; extra heads existed only during training. Each candidate was trained with 3 seeds for 40 epochs, keeping the epoch with the best dev macro-F1. A 10-seed confirmatory run and the test were reserved

for a candidate beating the original recipe by at least 0.5 dev points.

Candidate	Change from the original recipe	Dev macro-F1	vs C0
C0	Original recipe, our retraining	54.12	0.00
C1	Mean of four RoBERTa layers as the text input	54.05	−0.07
C2	Prototype contrastive loss (SPCL-style), with Fear vs Sadness, Anger, Surprise and Disgust vs Anger weighted ×2; loss weight 0.1 / 0.3 / 1.0	53.91 / 53.78 / 53.07	−0.22 / −0.34 / −1.05
C3	Auxiliary sentiment head on MELD's sentiment labels; loss weight 0.3 / 1.0	54.31 / 53.24	+0.19 / −0.89
C4	Learning rate 3e-5 instead of 1e-5	54.06	−0.06

Values are means over 3 seeds; the seed standard deviation is 0.5 to 1.1 points.

What it showed.

- **Our retraining reproduces the released models:** 54.12 dev macro-F1, against 54.04 for the three released checkpoints.
- **ECERC is not under-trained.** At 3× the learning rate it reaches the same dev peak by epoch 18 instead of 38, then overfits (mean of the last five epochs: 50.6).
- **Fear and Disgust stay at about 30 to 36 dev F1 under every variant,** and the stronger auxiliary losses hurt.

Conclusion. ECERC's rare-class ceiling is set by its input features and the data, not by its objective or training budget. Improving rare emotions needs better evidence about them.

8. Experiment 4: ECERC fused with a RoBERTa-large text expert

Adding a fine-tuned RoBERTa-large model produced the only rare-class gain on test: Fear F1 rose 3.40 points. Macro-F1 rose only 0.27 (95% CI −1.92 to +2.29), so the verdict is INCONCLUSIVE.

Idea. Experiments 1 to 3 pointed to ECERC's frozen features as the limit. A RoBERTa-large model fine-tuned end to end on the dialogue text scores about as well as ECERC (66.57 weighted F1 in TelME). It learns from raw words and context, so its errors may differ.

Setup.

- **Expert:** RoBERTa-large fine-tuned on MELD's training dialogues. The input is the past turns and the current one, each as "Speaker k: text", then the prompt "Now Speaker k feels `<mask>`". A linear head reads the mask token. It trained for 6 epochs with 3 seeds, keeping the best dev macro-F1 epoch.
- **Fusion:** score = (1 − w)·log p(ECERC) + w·log p(expert), with w chosen on dev from 0 to 1 in steps of 0.05. The 3 released checkpoints × 3 expert seeds give 9 pairs.
- **Gate:** the cross-fitted dev gain was +0.97, above the +0.5 gate, so the test ran once with w* = 0.5.

Test, mean of 9 pairs	Accuracy	Weighted F1	Macro-F1	Fear F1	Disgust F1
ECERC alone	67.32	66.46	50.87	26.12	31.43
ECERC + expert, w = 0.5	67.12	66.45	51.14	29.52	29.46
Expert alone	65.57	65.34	50.50	27.87	31.89

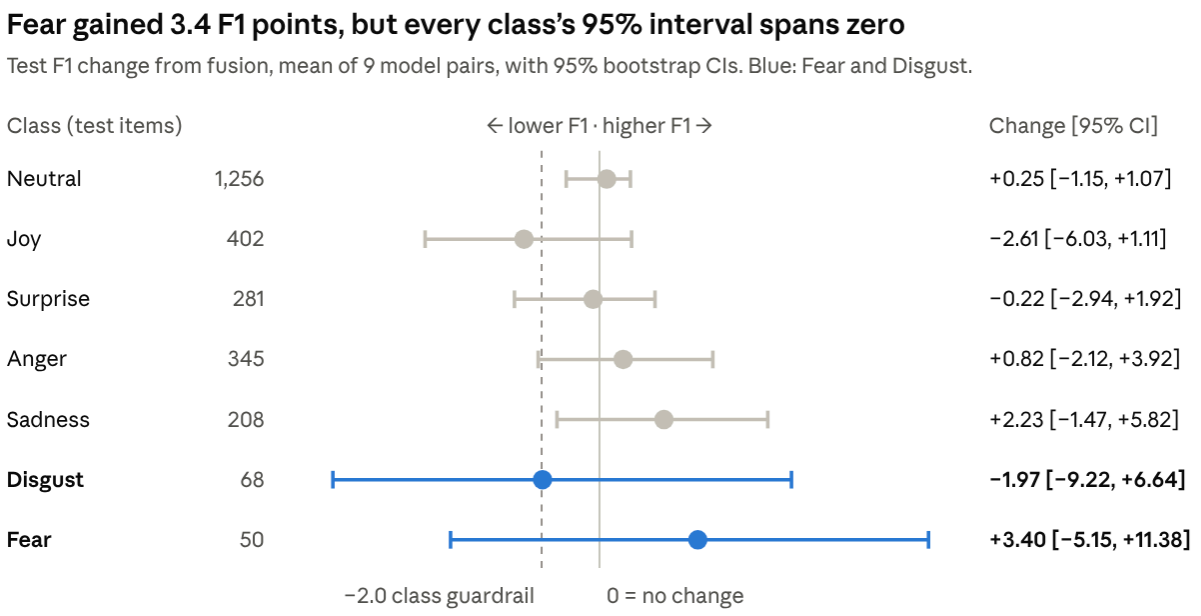
Against the criteria. Fear improved, and accuracy (−0.20) and weighted F1 (−0.01) stayed within their guardrails. But the gain was below +1.0 and its CI includes zero. Disgust fell 1.97 and Joy 2.61, breaking the 2.0-point class guardrail, and only 5 of 9 pairs improved.

Other analyses (not confirmatory). Choosing w and a logit adjustment τ together on dev (w = 0.2, τ = 0.1) gave +1.17 macro-F1, with a test-only CI of −0.01 to +2.36. That interval ignores selection noise, so it cannot overturn the verdict. Plain ensembles scored 50.31 (three ECERC models), 51.05 (three experts) and 51.42 (all fused pairs).

What it showed.

- **The expert sees rare-class evidence that ECERC misses.** It finds 31% of test Fear utterances against ECERC's 21%, and fusion lifts Fear, the paper's worst class, from 26.1 to 29.5 F1.
- **The gains are paid for elsewhere.** Disgust and Joy lose about as much as Fear and Sadness gain, so macro-F1 barely moves.
- **The test set cannot resolve effects this small.** Dev macro-F1 is nearly flat for w between 0.15 and 0.6, so the chosen w swings across bootstrap replicates. With 50 Fear and 68 Disgust test items, the smallest effect detectable 80% of the time is about 2.6 points.

The chart below shows each class's change with its CI.



Experiment 4 test results: ECERC + expert (w = 0.5) minus ECERC alone, mean over 9 model pairs; procedure-level dialogue bootstrap, 10,000 replicates.

With only 50 Fear test items, a single utterance moves Fear recall by 2 points, which is why the rare-class intervals are so wide.

9. Experiment 5: ECERC fused with a Qwen2.5-7B LLM expert

A fine-tuned 7-billion-parameter language model beat ECERC on overall dev scores but recognized Disgust far worse. Fusing it with ECERC lowered cross-fitted dev macro-F1 by 0.98 points, so the gate stopped the run before the test.

Idea. Fine-tuned large language models report some of the highest MELD scores (InstructERC: 69.15 weighted F1, against ECERC's 66.46). A stronger reader of the dialogue might add rare-class evidence beyond experiment 4's expert.

Setup.

- **Model:** Qwen2.5-7B-Instruct with QLoRA: a 4-bit base model, LoRA adapters (rank 16) on every attention and MLP projection, and a classification head.
- **Input:** a short instruction, as many past turns as fit, and the target utterance, capped at 256 tokens.
- **Training:** cross-entropy, 1 epoch, learning rate 1e-4, effective batch 16. Two seeds trained in parallel on two Kaggle T4 GPUs in 98 minutes, with a dev check after each third of the epoch.
- **Fusion, gate and test:** as in experiment 4, with 6 pairs. As this would have been the third test look, a Bonferroni-adjusted 98.33% CI was also planned.

Dev	Accuracy	Weighted F1	Macro-F1	Fear F1	Disgust F1	Sadness F1
ECERC, C1 / C2 / C3	67.0 / 66.6 / 66.6	65.9 / 65.6 / 65.4	54.4 / 53.5 / 54.2	30.8 / 32.3 / 30.0	42.1 / 35.9 / 43.2	44.2 / 44.6 / 43.8
LLM, seed 1 / seed 2	67.0 / 69.0	66.4 / 66.9	53.5 / 52.7	36.2 / 24.0	23.1 / 22.2	42.9 / 47.3

What it showed.

- **Better overall, worse on rare emotions.** The LLM leads on accuracy and weighted F1, which Neutral, Joy and Surprise dominate. It trails on macro-F1, mostly because of Disgust: 22 to 23 F1 against ECERC's 36 to 43.
- **The in-sample gain did not survive cross-fitting.** The best dev weight (w = 0.45) gave +0.83 on the dev data that chose it, but −0.98 out of fold.
- **Training was short and noisy.** Within one epoch, dev macro-F1 moved 40 → 54 → 51 (seed 1) and 45 → 47 → 53 (seed 2). Longer training might score higher, but nothing on dev suggests it would fix Disgust.
- **Possible contamination.** The LLM's pretraining data may include *Friends* transcripts. That would inflate the LLM's own scores, so it cannot explain the negative result.

Conclusion. A larger model trained with ordinary cross-entropy on MELD inherits the same imbalance. Fusing it with ECERC trades one class for another.

10. Why it happened

Limitation 2 resisted every fix because the rare-class errors come from the data and the input features, not from ECERC's decision rule or training. Five causes, each backed by our runs, explain the pattern.

Cause	Evidence	What it explains
Base rates swamp good rankings	ECERC ranks Fear above Neutral well (dev AUROC 0.95), but Neutral is 17× more common. Utterances near the boundary are therefore mostly Neutral: 19 extra Fear predictions gave 1 hit.	Why threshold shifts (exp. 1), class weights and oversampling trade precision for recall without raising F1.

Cause	Evidence	What it explains
Related emotions overlap, and labels are noisy	Fear is lost to Surprise, Sadness or Anger (38%) as often as to Neutral (33%), and is harder to separate from them (AUROC 0.84 to 0.87). MELD's annotator agreement is $\kappa = 0.43$.	Why no boundary or loss change separates them (exp. 1 and 3): many Fear items look like Surprise or Anger, and some may honestly be both.
The frozen features carry little rare-class signal	Probes find 24% of Fear even in unseen training dialogues, and ECERC's own training-set Fear recall is 31%. The same items fail in all three checkpoints (30 of 31 for Fear). A 3× learning rate reached the same peak.	Why changing ECERC's loss, inputs or optimizer (exp. 3) could not help: the information is missing from its input.
Every model trained on MELD's labels inherits the skew	The RoBERTa expert gained Fear (+3.40) and Sadness (+2.23) but lost Disgust (−1.97) and Joy (−2.61). The LLM beat ECERC on weighted F1 but scored 22 to 23 on Disgust, against 36 to 43.	Why fusing experts moves errors between classes instead of removing them (exp. 4 and 5).
The test set is too small to see small gains	MELD's test set has 50 Fear and 68 Disgust utterances. Their per-class CIs are about 16 points wide, and the smallest reliably detectable macro-F1 effect is about 2.6 points.	Why the +0.27 gain (and the secondary +1.17) cannot be confirmed, and why the earlier 3-seed study was inconclusive.

On the paper's explanation. The paper blames "excessive Neutral training samples". That is only half right: Neutral absorbs about a third of rare-class errors, but not as a bias a prior correction can remove (exp. 1), and related emotions absorb as many. Checkpoint selection by weighted F1 is not the cause either: selecting by macro-F1 gave the same dev score (54.12 against 54.04).

The one positive lever. A model that reads the raw dialogue found 31% of test Fear utterances, against ECERC's 21% (exp. 4). Better evidence, not better balancing, is what moved a rare class.

11. How it could be improved, beyond our resources

The routes left open change the evidence or the labels, not ECERC's decision rule. Each needs data, code or compute this project did not have.

Direction	Why it should help	What it needs that we lacked
Learn audio and visual features from the raw clips (e.g. wav2vec 2.0 or WavLM for speech, face and body-pose encoders for video)	Fear and Disgust often show in voice and face, but the provided visual features carry almost no label signal (probe macro-F1 14). A 2026 pose-aware MELD system reports Fear +9.19 and Disgust +6.07 F1 from added pose features.	MELD's raw video, a feature-extraction pipeline and multi-GPU training
Train the encoders end to end with an imbalance-aware or contrastive loss	Experiment 3 could not add information to frozen features, while experiment 4 showed end-to-end training finds more Fear. SPCL and MultiEMO's sample-weighted focal contrastive loss target confusable minority emotions.	ECERC's unreleased feature-extraction code and a large GPU budget
Relabel, or use soft labels for, rare and confusable utterances	Agreement is low (κ = 0.43) and the same items fail in every model. A 2026 re-annotation study of ERC utterances found strong agreement in only 35% of cases. Soft labels stop penalizing plausible answers.	Annotators, the raw clips and time
Add rare-class training data (augmentation, or transfer from other ERC corpora such as IEMOCAP and EmoryNLP)	Fear and Disgust have only 268 and 271 training utterances.	Label mapping across datasets, quality control and compute to retrain

Direction	Why it should help	What it needs that we lacked
Rebuild the text features with cross-fitting (out-of-fold features for training utterances)	It removes the +7.6-point train-side advantage found in experiment 2.	The unreleased feature-extraction code
Train the LLM expert longer, or a larger one (several epochs, InstructERC-style auxiliary tasks)	Our one-epoch run was noisy and under-trained; InstructERC reports 69.15 weighted F1 on MELD.	A100-class GPUs for days, beyond free T4 sessions
Evaluate on a larger or second test set, with macro-F1 as the primary metric	MELD's test set cannot resolve effects below about 2.6 macro-F1 points.	A new labeled test set

The headroom is real but modest. Even strong recent multimodal systems report only about 31 Fear and 43 Disgust F1 on MELD (MiSTER-E), against ECERC's 26 and 31.

12. Limitations of this study

Our conclusions hold for ECERC's released MELD models on MELD's test set. These caveats bound them.

- **Baseline models.** Experiments 1, 4 and 5 used the authors' three released checkpoints. The results describe those models, not the full spread of ECERC training seeds.
- **A small dev set.** Dev has only 40 Fear and 22 Disgust utterances, so every dev-based choice is noisy. A useful method could have been stopped by the gate by chance; cross-fitting reduces this risk but does not remove it.
- **Statistical power.** The test set detects effects of about 2.6 macro-F1 points for fusion (experiment 4), or 0.9 for a same-model comparison (experiment 1). Smaller real gains could go unseen.
- **Two test looks.** The test set was opened twice, in experiments 1 and 4. Each verdict uses its own pre-registered 95% rule; neither is a positive claim, so the repeated look cannot create a false success.
- **Limited training budgets.** Experiment 3 used 3 seeds per candidate, and the LLM trained for one epoch on T4 GPUs. Larger budgets might change experiment 5's numbers, though nothing suggests they would fix its Disgust deficit.
- **One dataset.** We studied MELD only; IEMOCAP was reserved for a positive MELD result, which never came.

- **Disclosed extras.** The LLM's pretraining data may include *Friends* transcripts. Candidate C3 used MELD's sentiment labels, which the original ECERC does not use.

13. Conclusion

Within ECERC's design, Limitation 2 cannot be fixed through the decision rule, the training objective or the training budget. This negative answer is backed by pre-registered tests with a sealed test set.

We can claim three things. It is not a threshold problem: the +1.0 target was excluded (experiment 1). The text features do not leak training labels strongly (experiment 2), and ECERC's training is not the bottleneck (experiment 3).

We cannot claim that fusion helps. Experiment 4's Fear gain is encouraging, but its macro-F1 CI includes zero, and Disgust and Joy fell. Experiment 5 shows a much larger model does not change this.

The lever left is better rare-class evidence: raw audio-visual signals, cleaner labels or more data. The protocol and notebooks here can test such features directly once they exist.

14. Reproducibility

Every run can be repeated from one notebook, and all code, rules and raw outputs live in the project repository under `research/limitation2/`. The original ECERC code in `MELD/` and `IEMOCAP/` is untouched.

Exp.	Notebook (in `notebooks/`)	Platform and run time	Outputs (in `results/`)
1	`ECERC_L2_posthoc_LA.ipynb`	Colab, 1 × T4, about 15 min	`run_colab_1/`
2	`ECERC_L2_feature_leak_check.ipynb`	Colab, CPU, about 10 min	`run_colab_leak_1/`
3	`ECERC_L2_pathB_representation.ipynb`	Kaggle, 1 × T4, 38 min	`run_kaggle_pathb_1/`
4	`ECERC_L2_expert_fusion.ipynb`	Kaggle, 1 × T4, about 2 h 17 min	`run_kaggle_expert_1/`
5	`ECERC_L2_llm_fusion.ipynb`	Kaggle, 2 × T4, about 1 h 40 min	`run_kaggle_llm_1/`

- **Self-contained notebooks.** Each embeds its analysis code from `src/` with a SHA-256 check. It downloads the data and checkpoints from the authors' official links and verifies their hashes against `MANIFEST.json`.
- **Rules.** `PROTOCOL.md` holds the pre-registered rules for all five experiments, each written before its run.
- **Decisions and results.** Each run folder holds the frozen config, the integrity checks (all passed), the dev selection, library versions and, for experiments 1 and 4, the single test result. `results/FINDINGS.md` interprets every run.
- **Original code.** The four MELD files used were copied verbatim to `ecerc_original/`, and their hashes match the originals.
- **Local tests.** `tests/smoke_test*.py` run each notebook end to end on synthetic data on a CPU. They fail if a test label is read before the test stage.
- **To rerun:** upload a notebook to Colab or Kaggle, select the GPU shown above, and run all cells. `README.md` gives the step-by-step instructions.

15. References

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